ECHO SHIFT

Power the path. Send your echo into the dark.

COSC 3026 Game Design I · Term Project Proposal · Instructor: Dr. Ratvinder Grewal

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Story

An experiment with time takes place in an underground laboratory facility, but everything goes wrong as the electricity is cut off and a creature from the dark comes out into the open. A scientist gets stuck underground and all she has is a generator and a recorder which will keep one trace of her actions.

Gameplay

Echo Shift is a single-player, top-down, real-time puzzle game. In the Echo Shift game, players can record their route to replay it as an echo, a ghost version of themselves. All the rooms are not powered by the generator, and players can stay safe only in the light. That is why the players record their route when the room is lit, move the power of the room to another one, and let the echo do the rest in darkness.

Unique selling points

• It is the echo who undertakes all dangerous actions, capable of exploring dark areas inaccessible to the player.

• Each level consists of a power budget and picking which area to keep dark is the challenge.

• One simple, deadly rule: a single step into the dark fails the level, so every route has to be

planned. There are no jump scares.

Intended audience

For young adults and teens age 13 and older who like to solve logical puzzles and like planning ahead more than fast action. The ESRB rating will be E10+ as the game includes fantasy violence but does not have any blood and gore.

How a level plays: a simple example

The power produced by the generator equals 60. Lights of the laboratory consume 30 and the elevator consume 50. Thus, both cannot operate at the same time. The switch which operates the elevator is located within the lab, but it can work only when there is power in the elevator.

1. Record: Having entered the lit corridor and pressed the switch, the player sees that it is not working since the elevator lacks power. After the player stopped recording, the recording device takes him back to the beginning of the route to the lit corridor.
2. Move the power: In the corridor, the player switches off the laboratory lights and switches on the elevator. No one ever enters the dark laboratory.
3. Send the echo: The echo travels the route in the dark laboratory and presses the switch. The creature cannot damage it. The elevator opens and the player moves to it in the lit corridor.

Initial thoughts on mechanics

Mechanic	How it works
Movement	The player moves around the level in real time using the keyboard.
Recording	The player starts recording and moves normally for a limited number of turns (about 12, to be tuned in playtesting). Stopping the recording rewinds the player to where they started.
The echo	The echo replays the recorded moves one turn at a time while the player makes new moves. It can press switches and stand on pressure plates. Only one echo exists at a time.
Power	Each level has a fixed amount of generator power. Every room's light and every machine cost some of it. The player switches things on and off at a power panel and can never go over the limit.
Light and the creature	Lit rooms are safe. The creature moves only through dark areas, one tile per turn. If it reaches the player, the level restarts. It cannot harm or block the echo.
Restarting	There are no lives. The player can restart a level at any time with no penalty.

Initial thoughts on interface

Controls

Key	Action
Arrow keys / WASD	Move one tile
Space	Wait one turn
Q	Start or stop recording
E	Use the power panel (when standing at it)
R	Restart the level
Esc	Pause menu (level select, sound on/off, quit)

On screen

- The level map, shown top-down on a tile grid.
- A power meter showing power used out of the total, with each room and machine listed as ON or OFF.
- A recording indicator and a turn counter, so the player knows how many recorded turns are left.

Look and sound

- Retro pixel art. The player is amber, the echo is a glowing cyan outline, and dark rooms are almost black with only the creature's eyes visible.
- Short sounds for recording, switching power and opening doors, and a low warning sound when the creature is close.

Similar games and features we plan to use

GAME	FEATURE WE WANT TO USE
CHRONOTRON (SCARYBUG GAMES, 2008)	Solving puzzles by working together with recordings of yourself
THINKING WITH TIME MACHINE (STRIDEMANN, 2014)	A recorded copy of the player acting as a puzzle partner
FIVE NIGHTS AT FREDDY'S (S. CAWTHON, 2014)	Limited power spent on lights, with danger in the dark (we will keep the tension but leave out jump scares)
FTL: FASTER THAN LIGHT (SUBSET GAMES, 2012)	Sharing a small amount of power between systems

Platform and technology

Our game is for Windows PC (10/11), local only and played with a keyboard. We intend to implement it in Python, using the lightweight library Pyxel (a Windows library for retro style games, that manages the drawing, the keyboard and the sound). As this library has no game logic, we will have implemented all rules of the puzzles, giving us control of the game design. We will use Git/Github to share code within the group, and will distribute the finished game as a Windows program.

Development plan

Stage	Goal
Proposal	Present the idea and collect feedback from the class and the instructor.
Mid-term	A playable demo with the core mechanics (recording, power and the creature) in the first few levels.
Final	About six finished levels, packaged as a Windows program.

References

Cawthon, S. (2014). *Five Nights at Freddy's* [Video game]. Scott Cawthon.

Kitao, T. (n.d.). *Pyxel user guide*. GitHub.

https://github.com/kitao/pyxel/blob/main/docs/user-guide.md Scarybug Games.

(2008). *Chronotron* [Video game]. Kongregate.

https://www.kongregate.com/en/games/scarybug/chronotron

Stridemann. (2014). *Thinking with Time Machine* [Video game mod]. Steam.

https://store.steampowered.com/app/286080/ Subset Games. (2012). *FTL: Faster than light* [Video game]. Subset Games.